

Navigating Advanced AI: Opportunities, Risks, and the Role of Philanthropy

Short Summary

This report explores the vital role that philanthropists can play in ensuring beneficial outcomes from advanced AI systems. The report outlines the key capabilities and risks posed by AI systems and highlights key areas where philanthropists can strategically make a substantial impact, such as boosting global resilience, supporting technical research, facilitating international coordination, and building government capacity to regulate AI effectively.

Executive Summary

Artificial Intelligence (AI) is advancing at a rapid pace, offering profound potential to improve health, wealth and wellbeing across the globe. However, these advances come with significant risks, especially as AI systems grow in capability and autonomy. In this report, we outline how philanthropy can effectively mitigate these risks and ensure beneficial outcomes from advanced AI. This report is structured into three sections:

1. First, we summarize the [current state of AI](#), evaluating both its capabilities and risks.

We analyze how systems are becoming more [capable](#), increasing both in generality and performance, with ever-greater societal applications. Further, we point to how the *rate* of improvement is increasing, indicating that **AI development is on an exponential trajectory**. We highlight how this trendline may lead to **fully general, human-level AI systems**, which could have transformative implications.

We then study the risks associated with advanced AI systems, noting that AI could (among other harms) cause mass unemployment, proliferate misinformation, exacerbate discrimination, enable rogue actors to cause immense harm, and act in unpredictable, uncontrollable ways with catastrophic consequences.

2. We then consider [the role that philanthropy](#) can play in mitigating these threats. We note that **big tech companies are racing to build advanced AI systems without adequate investment in dealing with the safety, security or ethical challenges** that they pose on wider society.

Further, we note that **democratic governments are not prepared** to mitigate these risks, due to a lack of expertise and a vulnerability to lobbying by big tech companies.

We argue that given the limitations of the private and public sector to adequately navigate these challenges, it is imperative that civil society– and in particular, philanthropists – step up to the challenge. We highlight that philanthropists have previously moved the needle on major issues in emerging technologies, but note that the field of AI is highly imbalanced - with **\$250 invested in making AI systems more powerful for every \$1 spent on ensuring they are safe**. Further, we observe that the funding that is dedicated towards AI risks could be more effectively spent.

3. Finally, we recommend that philanthropists focused on advanced AI should concentrate their efforts on four key focus areas:
 1. **Boosting global resilience**, to prepare for worst-case scenarios where preventative measures fail.
 2. **Supporting technical research**, particularly on open challenges overlooked by AI companies.
 3. **Facilitating coordination**, both between great powers and leading frontier AI companies.
 4. **Building government capacity**, equipping policymakers to build out the regulations required to meet the scale of the challenges we face.

We conclude that there are a wide range of actions that funders can take to ensure advanced AI is safely and beneficially developed and deployed. We argue that meaningful progress on AI **can** be made, and **must** be made, through ambitious, strategic and courageous intervention from philanthropists.

Introduction: What does success look like?

On the average day, about [16,000 children die](#) before they reach their fifteenth birthday. Often, such tragedies occur due to diseases which are easily treatable. The majority of the global population lives on [less than \\$10 a day](#) – whilst [the top 1% of earners take home one fifth of all global wealth](#). Meanwhile, hundreds of millions of animals are killed every day, [the vast majority](#) of whom spend their whole lives in factory farms. And average [global temperatures have spiked](#) over the past decades, set to cause significant deaths and destruction for billions.

Advanced AI systems – if safely developed and cautiously deployed– could make progress on these problems. Economic growth could dramatically rise, lifting millions out of poverty. Advanced models of human biology could generate rapid medical innovation, eliminating most human disease; whilst technological breakthroughs in energy production via accelerated R&D could allow for dramatic improvements to reduce climate change. If well governed, these positives will not just be for the benefit of a small minority, but for all. A profound improvement in the human condition would occur. And this is the *lower bound* of what is possible.

However, progress is **only** possible if we first navigate the risks. As such, we believe the greatest impact can be had by **minimizing the risks, in order to maximize the opportunities**. It is precisely **because** of AI's potential that we must be attentive to its dangers.

The current state of AI

The past few years have seen an explosion in interest and investment in AI. In late 2022, ChatGPT became the fastest growing consumer application in history, reaching [100 million users](#) in just two months. Broader investment in Generative AI has dramatically risen, reaching [\\$25 billion](#) in 2023. Below, we explore the underlying trends driving AI's dramatic rise, consider where it may take us, and analyze the societal risks it poses.

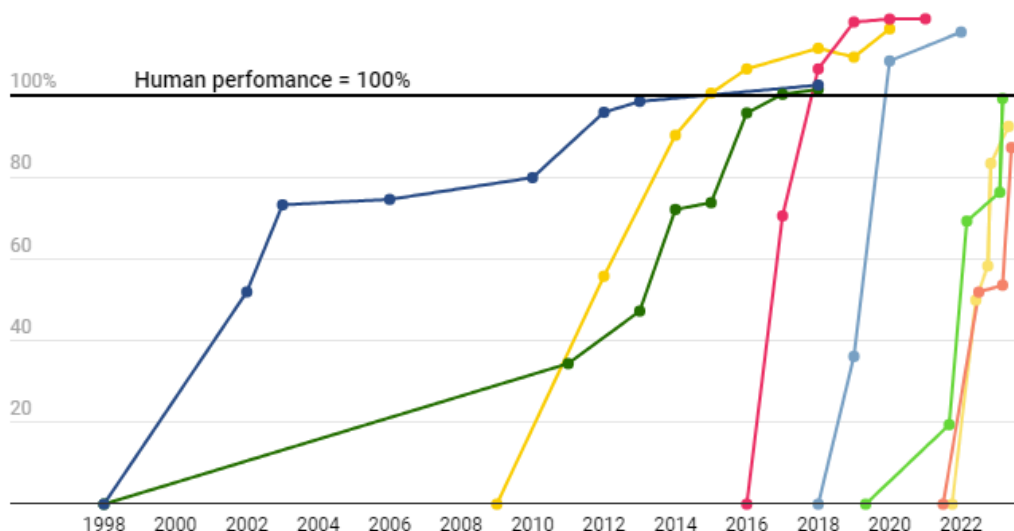
Capabilities

The last couple of years have seen an [extraordinary increase](#) in the performance of AI models. Indeed, across many tasks, AI systems now perform on a comparable level to humans, and the rate at which they are improving is increasing.

AI has surpassed humans at a number of tasks and the rate at which humans are being surpassed at new tasks is increasing

State-of-the-art AI performance on benchmarks, relative to human performance

● Handwriting recognition ● Speech recognition ● Image recognition ● Reading comprehension
● Language understanding ● Common sense completion ● Grade school math ● Code generation



For each benchmark, the maximally performing baseline reported in the benchmark paper is taken as the "starting point", which is set at 0%. Human performance number is set at 100%. Handwriting recognition = MNIST, Language understanding = GLUE, Image recognition = ImageNet, Reading comprehension = SQuAD 1.1, Reading comprehension = SQuAD 2.0, Speech recognition = Switchboard, Grade school math = GSK8k, Common sense completion = HellaSwag, Code generation = HumanEval.

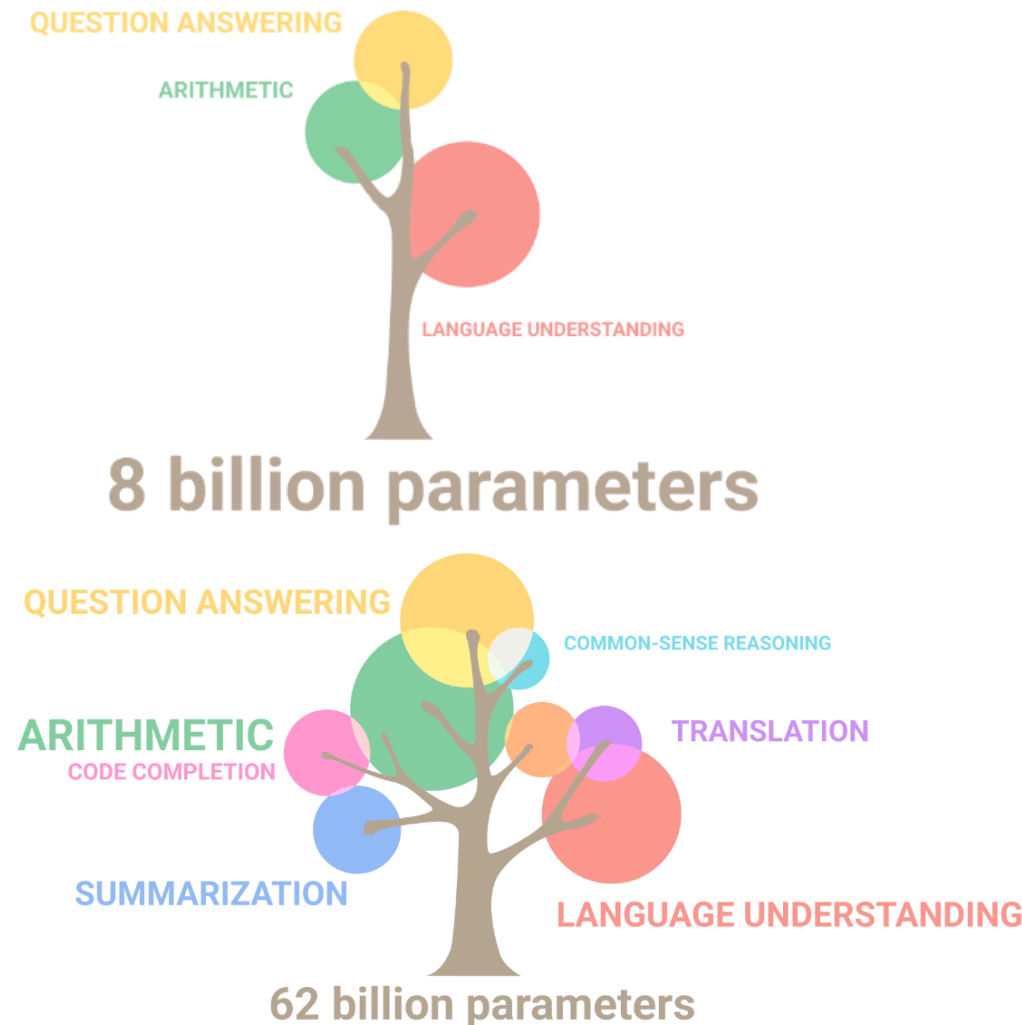
Chart: Will Henshall for TIME • Source: [ContextualAI](#)

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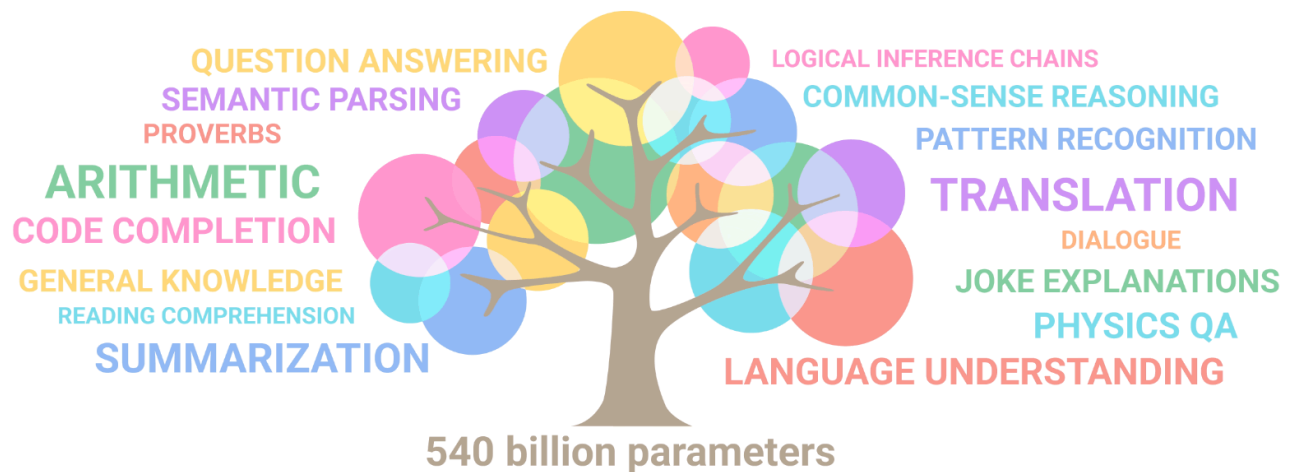


These rapid advances in performance have been primarily driven by the training of larger and larger AI models.¹ Indeed, the size of models at the frontier has [dramatically increased](#) over the last few decades, with the most advanced systems doubling in size every 10 months.

The below images illustrate how as these models grow in size (measured by number of parameters), they unlock new capabilities, as well as improving in existing domains.



¹ Concretely, “model size” refers to the amount of computation (“compute”) involved to train a model (typically measured in “[Floating-point Operations](#)”)



Source: [Google PaLM \(2022\)](#). Branches of the tree represent different abilities, whilst the size of the circles represents performance on this ability.

If AI systems continue to improve across performance and generality, then they may ultimately reach - and then surpass - human capabilities. If current trends hold, then highly advanced, [general AI](#) could plausibly be developed and deployed at some point in the next few decades, or even the next few years. Such a prospect offers both profound opportunities and deeply concerning risks.

Risks

Whilst advanced AIs may be highly beneficial, they are not without danger. AI systems could threaten worker livelihoods, spread misinformation, exacerbate discrimination, enable bad actors to cause harm, and act in unpredictable, uncontrollable ways.² There is a full spectrum of risks, from those that society is already confronting today to those that could pose catastrophic risks in the not-too-distant future.

Labor Market Disruption

Advanced AI systems are set to [significantly disrupt](#) the workforce, with the potential for both positive and negative impacts. On the one hand, AI may automate many laborious tasks, freeing up workers to pursue more valuable projects. On the other hand, entire jobs may be automated, with significant costs to the vast majority of workers whilst a small minority see extraordinary economic gains.

Degradation of the information environment

Frontier AI systems can already generate realistic [images](#), [videos](#) and [text](#). Such systems have the capability to create and spread false or misleading information, with the potential to cause harm. As AI capabilities advance, they may worsen issues such as [violence](#), [online media credibility](#), [public health crises](#), and [political partisanship](#).³ In future, such harms could reach

² Note the threats covered here are non-exhaustive

³ [How to prepare for the deluge of generative AI on social media](#). Kapoor and Narayanan, 2023



the level of undermining overall [epistemic security](#) - in the limit forming a risk factor for [catastrophic outcomes](#).

Bias, Fairness and Representational Harms

AI systems may perpetuate existing biases, unfairness and discrimination. This is because AI systems reflect the data they are trained on, and as such they [imitate biases](#) present in the training data - the same biases which can be found in wider society. Due to the [opaque nature](#) of current AI systems, biases are often deeply embedded and hard to remove. Thus, AI systems can have long-lasting impacts on [unfairness and discrimination](#) in society.

Misuse

AI systems could be misused by bad actors with catastrophic effect. For instance, bad actors may be able to use AI systems which aids in the development of novel high-lethality viruses, or in writing malicious software capable of disrupting critical infrastructure - which in turn may threaten lives or livelihoods on a large scale. As AI systems become both more powerful and more widely diffused (for instance, via open-source release), the number of actors accessing these capabilities will increase. With lower barriers to entry, the likelihood of such an event occurring increases.

Loss of Control

Advanced AI systems may be highly unpredictable and uncontrollable, making effective oversight challenging.

As AI systems continue to develop, they may eventually become capable of [exceeding human performance across a wide variety of domains](#). Simultaneously, such systems may be deployed to [independently pursue long-range, complex goals](#) (due to the economic advantages of doing so). In such a world, it will be hard to judge if AI systems are pursuing the originally intended outcome. Instead, they may pursue a [proxy goal](#) which only superficially achieves the goal - thus generating a misalignment. Worse, this misalignment may be [obscured from](#) the individuals responsible for overseeing these AI systems. If AI systems do become highly capable and goal-oriented, whilst pursuing goals which were not intended, human oversight will become impossible. In such a setting, powerful AI systems operating on their own could lead us into disaster. Whilst the likelihood of this risk materializing is a controversial topic, we believe the potential harms are great enough to warrant significant philanthropic attention.

Overall, there is a broad consensus that mitigating risks from advanced AI systems - up to and including extinction risks - warrants substantial attention, [alongside other societal-scale risks such as pandemics and nuclear war](#).



The role for philanthropy in positively steering advanced AI

Philanthropists can play an active role in ensuring beneficial outcomes from advanced AI by boosting global resilience, facilitating international coordination, building government capacity and accelerating technical research. Smart philanthropists have a vital role in supporting these efforts, stepping up to act where big tech and governments are missing the mark.

Big tech companies are incentivized to act irresponsibly

Whilst AI companies are set to earn enormous profits from developing powerful AI systems, the costs these systems impose are borne by society at large. These costs are [negative externalities](#), like those imposed on the public by chemical companies that pollute rivers, or large banks whose failure poses systemic risks.

Further, as companies engage in fierce competition to build AI systems, they are more inclined to cut corners in a race to the bottom. In such a race, even well-meaning companies will have fewer and fewer resources dedicated to tackling the harms and threats their systems create. Of course, AI firms may take some action to mitigate risks from their products⁴ - but there are well-studied [reasons](#) to suspect they will underinvest in such safety measures.

Governments are unprepared for AI risks

When the private sector fails to address societal challenges, it is the government's role to intervene. In the case of AI risk, however, there are several reasons to suspect that such intervention will be inadequate at dealing with the threats posed:

- *Political short-termism*: In democratic governments, politicians need to satisfy voter demands in order to be elected. Electoral pressures can lead to the prioritization of short-term results over sound long-term policy.
- *Limited technical capacity*: Governments lack the technical expertise required to understand - and therefore effectively regulate - AI systems.⁵
- *Regulatory capture*: Due in part to a lack of capacity, governments may be targets for capture by big tech companies, who are developing AI systems (and therefore able to push for policies in their own self-interest).

Governments have an important role to play - but there are compelling reasons to think they may be ill equipped. Philanthropists can ensure they get the support and advice they need.

⁴ After all, doing so is in their self interest, since it will help them become more marketable and profitable.

⁵ Note that government's have started to make progress in this direction - for example by establishing "AI safety institutes" in the [UK](#) and [US](#). However, these institutions remain embryonic, and broader technical capacity remains low.



Philanthropists have already played a key role in the development of emerging technologies, and can do so again

Given the scale and complexity of challenges in AI, it can be difficult to see how one can make a positive impact. Yet smart funders have made a vital difference on dangerous emerging technologies before, and can do so again.

In the 1980s, the [Carnegie](#) and [MacArthur](#) Foundations raised the alarm around “[loose nukes](#)” - the risk that nuclear weapons could fall into dangerous hands following the collapse of the Soviet Union. Further still, they funded the development of the [Cooperative Threat Reduction Program](#) to deal with the issue, which was passed by the US Congress. As a result, thousands of nuclear warheads were decommissioned. It is very possible that, without their intervention, the US government would not have dealt with the “loose nukes” problem.

The example set by Carnegie and MacArthur illustrates that philanthropists can play a critical role in mitigating large scale risks from emerging technology. We believe that the same can be true when it comes to risks from advanced AI, if philanthropists are thoughtful and strategic with their giving.

For every \$1 spent on ensuring AI systems are safe, \$250 is invested in making them more powerful.

When both the private and public sectors fail to address a societal issue, there is often a role for philanthropy to step in. However, in the case of mitigating AI risks, funding from philanthropy is concerningly low. [McAleese \(2023\)](#) estimates that AI safety received around \$100 million / year in philanthropic support over the past few years. For comparison, approximately \$3 billion is spent by philanthropists on climate change - approximately 30x more than on AI risk mitigation.⁶

Meanwhile, around \$60 billion was invested in AI developments in 2023,⁷ with around \$24 billion at the frontier.⁸ As such, for every \$1 spent on ensuring AI systems are safe, ~\$250 is invested in making them more capable.

Existing philanthropy could be spent more effectively

There are several reasons to suspect that the existing philanthropic landscape on AI risks could be improved.

⁶ [ClimateWorks, 2022](#)

⁷ \$45 billion invested in AI as of September 29 2023 ([State of AI Report](#) 2023, Slide 109). Assume an extra \$15 billion for the final quarter of 2023.

⁸ Investment in GenAI, representing the frontier. [State of AI Report](#) 2023, Slide 116. Assume an extra 25% for the final quarter of 2023.

1. *High centralization*: At least 80% of funding tackling risks from advanced AI comes from two funders.⁹ With such a high concentration, the collapse of either funder would have a significant cost. Further, there are limited competitive pressures on funders to find promising AI risk funding opportunities.
2. *Limited capacity*: AI risk grantmakers lack capacity to evaluate potential grants, slowing the deployment of funding into the field.¹⁰
3. *Conflicts of Interest*: Existing funders tackling risks from advanced AI systems have close connections to both AI firms and AI safety organizations. Such connections may be a potential conflict of interest which could impair the impartiality of funding decisions.
4. *Geographic bias*: Existing funders are almost entirely based and focused on the US. Whilst the United States is potentially of highest importance in AI, specialization in this geography may come at the exclusion of other highly effective but neglected opportunities in other countries.
5. *Anti-governmental bias*: Most funders targeting *advanced AI* have historically been concentrated on technical over policy solutions.¹¹ But governments are playing an increasing role in regulating AI risks.

Priorities for impact-focused philanthropists

In this section, we outline a grantmaking strategy informed by all the preceding sections.¹² We recommend:

1. **Boosting global resilience**, to prepare for worst-case scenarios where preventative measures fail.
2. **Supporting technical research**, particularly on open challenges overlooked by AI companies.
3. **Facilitating coordination**, both between great powers and leading frontier AI companies.

⁹ Open Philanthropy has donated over \$370m towards AI between 2015 and May 2024 - see [Potential Risks from Advanced Artificial Intelligence | Open Philanthropy](#). The SFF has donated about \$30 million since 2019. Other funders - such as the [Long-Term Future Fund at EA Funds](#) and [Lightspeed Grants](#) - have historically received most funding from Open Philanthropy or SFF:

- Lightspeed: “Our primary funder for this round is Jaan Tallinn [founder of SFF]”
- ~50% of LTFF’s funding in 2022 came from Open Philanthropy or other institutional donors (mainly SFF) - [source](#)

¹⁰ See previous footnote, which notes that the largest funder of AI safety has historically not had more than 3 FTE employees focused on grantmaking

¹¹ This is a weakly held intuition about the historically contingent nature of how the field of AI safety has progressed. Specifically, it appears that concerns about advanced AI - especially misalignment - grew out of a technology-based culture in Silicon Valley, rather than a policy-based culture in D.C. Some (e.g. [Critch 2021](#)) have highlighted that this has historically caused a blindspot around political/policy over technical solutions. It is worth noting that a significant and growing fraction of resources and attention are now devoted towards policy. Nonetheless, there may be some interventions still missed from this neglect.

¹² Note that these recommendations are based on the state of the AI risk field in early 2024, and so may not represent our approach in future years.

4. **Building government capacity**, equipping policymakers to build out the regulations required to meet the scale of the challenges we face.

Global resilience

We recommend interventions which plan for worst-case scenarios - that is, interventions which are effective when preventative measures fail to prevent AI threats emerging. For concreteness, we outline some potential interventions which boost resilience against AI risks.

- **Developing contingency plans:** Ensure there are clear plans and protocols in the event that an AI system poses an unacceptably high level of risk.¹³ Such planning could be analogous to planning in other fields, such as pandemic preparedness or nuclear wargaming.
- **Robust shutdown mechanisms:** Invest in infrastructure and planning to make it easier to close down AI systems in scenarios where they pose unacceptably high levels of risk.
- **Increase psychological defenses:** Beyond physical infrastructure protections, invest in preparing humans for defenses against persuasion and manipulation from AI systems.¹⁴
- **Tackle existing threats:** Without advanced AI, there are already threats of deep concern - such as pandemics and nuclear war. Plausibly, advanced AI systems may be used to cause harm through these existing threat pathways. Thus, for the sake of increasing resilience to AI risks, we might also wish to tackle other threats.

There are several reasons to suspect resilience is a promising approach:

- **Resilience is threat-agnostic:** Interventions which increase global resilience provide defenses against a multitude of threats. For example, restricting the use of AI in high stakes domains (such as nuclear or biological weapons) may be beneficial both for reducing the exacerbation of existing risks, in addition to tackling novel AI threats.
- **Investments in resilience offset other funders' focus areas:** As noted [previously](#), the market, governments and existing philanthropists have various biases, causing them to invest in a narrow set of priorities - such as particular governance mechanisms and research directions. Thus, philanthropists should prepare for worlds in which these priorities fail to prevent risks materializing.

Novel research approaches to tackle unsolved safety challenges

We recommend funding underexplored research approaches to tackle key challenges in building safe AI systems. Specifically, we focus on issues related to misalignment, and in

¹³ Whilst much focus has been placed on *evaluating* dangerous capabilities in AI systems, much less detail has been clarified on what ought to happen in the event that AI systems *are* dangerous.

¹⁴ Since the persuasiveness of AI systems is already increasing ([Salvi et al 2024](#); [Durmus et al 2024](#)), and may represent a weak node in the pathway towards several AI risk scenarios



particular “inner misalignment”,¹⁵ as we see this as a critical yet unsolved technical challenge. We believe this challenge is worth prioritizing for three reasons:

Advanced AI systems could be difficult to align and control, with potentially catastrophic consequences.

Advanced AI systems may be misaligned - that is, they have goals which differ from their users.¹⁶ If these future systems exceed human capabilities across a wide range of domains, then catastrophic harm to society could result if these systems pursue their own goals. Thus, alignment failures could pose some of the largest expected damage.

AI companies are not on track to solve key alignment problems - nor are they sufficiently incentivized to do so.

Some AI companies - such as [OpenAI](#), [Anthropic](#) and [Google DeepMind](#) - have made research efforts on AI safety challenges. However, there has recently been [backsliding](#) on those commitments, with some major labs leaving their safety teams poorly resourced. Even in cases where safety research has been supported, it has often also helped to only [superficially](#) make products more safe, prioritizing profitability over addressing risks more seriously.¹⁷ This leaves a large number of unsolved open problems in alignment and controllability.¹⁸ Pursuit of these research agendas can help to offset gaps in the existing research directions of AI labs.

Mechanisms for coordination between global powers, and between frontier AI companies

Finally, we recommend efforts to facilitate coordination between major actors in the development of advanced AI. Specifically, we are looking at:

1. **International coordination efforts**, with a particular focus on critical relationships such as US-China cooperation.
2. **Corporate governance action** to prevent a “race to the bottom” between frontier AI companies.

¹⁵ In brief, this is the challenge of ensuring that the objective we *specify* for an AI system is the objective that the AI system actually *learns* (rather than a correlated or orthogonal terminal goal), which incentivizes the system to [instrumentally converge](#) on power-seeking.

¹⁶ AI systems may develop goals which differ from user’s intentions due to reasons related to *goal misgeneralization* ([DeepMind 2020](#)), *deceptive alignment* ([Karnofsky 2023](#)), or a [Sharp Left Turn](#), among other reasons.

¹⁷ For concreteness, take the example of Reinforcement Learning from Human Feedback (RLHF). RLHF helps to “outer align” large language models to human preferences. Outer alignment of this variety also helps to make AI models more capable (and thus profitable). RLHF is thus a technique where **the correlation between alignment research and capabilities is high**. In future, we should expect similar alignment research (for instance [scaleable oversight](#) approaches like [Constitutional AI](#), [Reinforcement Learning from AI Feedback](#), and similar) to be funded by AI companies, given similar commercial incentives.

¹⁸ We are particularly excited about research agendas related to [AI control](#), [model organisms](#), [alignment theory](#), [robustness](#), and other directions which a) are uncorrelated with frontier labs, and b) are effective in worlds where alignment is “hard”.



Coordination mechanisms could take many forms, including treaties, statements and letters to signal agreement;¹⁹ the shared development, implementation and enforcement of regulations;²⁰ or dialogues,²¹ among other proposals. We think coordination is vital for two key reasons:

AI risks are primarily driven by coordination failures

Across our analysis of the [risks](#), it is rarely the case that a single actor unilaterally causes a disaster; instead, threats are often driven by multipolar dynamics between actors. This holds across several threats:

1. *Societal Harms*: Without intervention, AI development could cause a small number of actors to gain vast amounts of wealth and power, at the expense of wider society. Joint commitments between relevant parties are required *before* powerful AI systems are developed, such that one party is not able to later accumulate significant power or wealth.
2. *Misuse*: A key driver of misuse risks is from the *mass proliferation of dangerous capabilities*. Coordination amongst key players can help to prevent such a mass proliferation occurring.
3. *Loss of Control*: When AI companies are racing to deploy AI models, they will have less time to act cautiously and invest in appropriate safety measures.²² A similar dynamic can occur between nation states, potentially leading to a “race to the bottom” on safety measures.

Coordination is likely to become more challenging over time.

Coordination issues are further compounded by the ever-decreasing cost of building AI systems. In addition to AI capabilities advancing at the frontier, there are ever-increasing actors developing powerful AI systems (for instance via Open Source). Thus, it is imperative to ensure coordination between key existing actors, before global cooperation becomes infeasible.

¹⁹ Previous examples include [The Bletchley Declaration](#), [Statement on AI Risk](#) and [Pause Giant AI Experiments: An Open Letter](#)

²⁰ Example proposals include an International AI Agency ([IAIA](#)), with enforcement proposals including [Mittelstadt \(2021\)](#) and [Shavit \(2023\)](#)

²¹ For example, [track I](#) and [track II](#) dialogues

²² This dilemma can occur even if all actors would prefer to slow down and invest in safety. If an AI lab merely *perceives* that others are acting less cautiously, then they may feel obligated to deploy their own AI system first (even if they cannot guarantee their own system is safe). Indeed, the CEOs of [OpenAI](#), [Google DeepMind](#) and [Anthropic](#) have all signaled that AI development could lead to catastrophe - thus suggesting that failure is more likely driven by their failure to coordinate, rather than lack of awareness or concern from a single actor.

Boosting government capacity to understand emerging technology

Despite the prime importance of governments on mitigating the risks from AI, there remain a wide range of unsolved policy questions. Further, governments and their bureaucracies are blighted by [gridlock](#) and [decay](#), leaving them ill-equipped to effectively answer them.

Therefore, we recommend **increasing governments' capacity to govern AI**. This capacity could be boosted via a range of interventions, such as **supporting top experts inside government**; providing sound **external policy advice** and recommendations; or **advocacy** to commit governments towards effective safety measure in policy decisions.

We believe support for governments should be funded for four reasons:

Governments possess unique powers to mitigate AI

Philanthropists alone cannot mitigate AI risks - national governments hold unique powers, such as the ability to implement legally-binding policies, and to safeguard national security. This power means that in order to introduce many of the most essential interventions, philanthropists must engage with governments.

Government resources can be leveraged towards high impact opportunities.

As well as possessing unique powers, governments possess significant financial resources. There is evidence that governments are keen to direct some resources towards AI, including towards its potential risks.²³ However, such resources may not be strategically directed. Thus, one strategy is to [leverage](#) resources in beneficial directions, through advocacy and engagement with government by third parties.

Governments are inadequately equipped to tackle AI risks

Governments sometimes lack the necessary expertise to mitigate AI risks. This leaves them liable to lobbying by big tech companies, which possess significant resources to push for policies friendly to their commercial interests. To combat this, philanthropists have an important role, both in supporting civil society organizations, and in upskilling government officials on key policy matters.

Conclusion

Advanced AI systems could fundamentally transform civilization, on a scale greater than any previous technology. Such technologies are likely to be developed within the lifetime of many people on Earth today - and may even occur within this decade.

This report has sought to help navigate what is a highly complex, rapidly changing landscape, with a focus on how best to reduce the risks. Our overarching takeaway is that, despite the complexity, **these challenges are not insurmountable**. There are a wide range of actions one

²³ See the [AI Safety Institute](#) in the UK, or its [equivalent in the US](#), both of which have received resource

can take to reduce the risks - from technical research ensuring AI systems are controllable, to policy efforts ensuring that governments are equipped and coordinated to handle the risks. We believe meaningful progress **can** be made, and **must** be made, through ambitious, strategic and courageous intervention from philanthropists.

